Proposal: Enhancement of the BIMS Mobile Application

To: National Building Review Board (NBRB)

Project: Enhancement of the Building Industry Management System (BIMS) Mobile Application

Date: 19th February 2026

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Revision: 1.1

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1. Executive Summary

This proposal outlines a robust technical approach to enhancing the BIMS Mobile Application for the National Building Review Board (NBRB). By utilizing **Flutter**, we will deliver a high-performance, cross-platform solution (Android and iOS) that extends BIMS services to mobile users, enabling citizen self-service, field inspections, and real-time notifications. Our solution focuses on security, offline reliability, and seamless integration with the existing BIMS backend via RESTful APIs.

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2. Technical Methodology & Flutter Implementation

We will employ **Flutter** to ensure a single codebase delivers native-grade performance on both platforms, meeting the SRS requirement for an app launch time of 5 seconds.

- **Architecture**: We will use the **BLoC (Business Logic Component)** pattern to separate the UI from complex business logic, ensuring scalability.
- **API Integration**: Communication with the BIMS backend will be handled via the **Dio** package, featuring interceptors for **JWT-based authentication** and HTTPS/TLS encryption.
- **Local Persistence**: To handle intermittent connectivity, **Hive or Sqflite (based on confirmation of nosql or relational database)** will be used for local caching of inspection data and permit details.

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3. Detailed Functional Implementation

3.1. Notification Architecture (Firebase Cloud Messaging)

To meet the requirement for **real-time notifications**, we will implement **Firebase Cloud Messaging (FCM)** using a scalable enterprise architecture.

- **Implementation Strategy**:
    - **Packages**: We will use firebase_core and firebase_messaging for the core integration, along with flutter_local_notifications to handle foreground alerts without an internet connection.
    - **Token Management**: Upon app launch, the system will retrieve a unique **FCM device token**. This token will be securely sent to the BIMS backend

to allow for targeted notifications (e.g., status changes on a specific permit application).

- **Lifecycle Handling**:
  - **Foreground**: Alerts will be displayed as a custom in-app banner using FirebaseMessaging.onMessage.
  - **Background/Terminated**: We will implement a top-level _firebaseMessagingBackgroundHandler to process incoming messages even when the app is closed.
- **Actionable Notifications**: Clicking a notification will trigger onMessageOpenedApp, deep-linking the user directly to the relevant application status or messaging screen.

3.2. Professional & Citizen Workflow

- **Document Management**: Using *file_picker* and *image_picker*, professionals can upload or replace high-resolution drawings directly from their mobile device.
- **Status Tracking**: Users will receive real-time updates on application status changes via **Firebase Cloud Messaging (FCM)**.
- **Financials**: The app will allow for the regeneration of expired PRNs and retrying of indeterminate payment statuses.

3.2. Geo-Tagging & Field Evidence

To ensure the integrity of field inspections and whistle-blowing reports, we will implement a multi-layered geo-tagging engine:

- **GPS Capture**: Using the geolocator package to capture site coordinates with high accuracy.
- **Metadata Embedding**: Programmatically writing GPS coordinates (Latitude/Longitude) into the **EXIF data** of inspection photos.
- **Visual Watermarking**: Overlaying a non-removable watermark containing the **Timestamp** and **BIMS Application ID** onto captured images.

3.3. Authentication & Security

- **Identity Verification**: NIN-based verification for individuals and BRN/TIN for entities.

- **Multi-Factor Auth**: Implementation of OTP-based 2FA via SMS and biometric (Fingerprint/Face ID) login. local_auth to provide "Device-level authentication support" as required by the SRS
- **Permit Validation**: A built-in QR code and Serial Number scanner for real-time validation of building permits.

3.4. Offline-First Synchronization Architecture

To handle intermittent connectivity during field inspections, we will implement a **Repository Pattern** with a local-first philosophy.

- **Local Database**: We will use **Sqflite** or **Hive** to store all inspection data and application history on-device.
- **Durable Queue**: Every user action (like submitting a photo or completing a checklist) will be written to a persistent **Sync Queue**.
- **Background Sync Engine**: A background process will poll the queue and attempt to push data to the BIMS REST APIs when a network is detected.
- **Conflict Resolution**: If the same permit is updated on both the web and mobile, we will use server-authoritative timestamps to resolve conflicts.

3.5. Detailed UI/UX Screen Map

The application will consist of approximately **18–22 unique screens** across three distinct user roles (Citizens, Professionals, and Building Control Officers) to ensure a comprehensive building control workflow.

Core Common Screens (5 Screens)

1. **Splash/Onboarding**: Branded entry with safe-building messaging.
2. **Authentication Hub**: Login via username/password, OTP-based 2FA, or Biometric (Fingerprint/Face ID).
3. **Unified Registration**: NIN-based verification for individuals or BRN/TIN for entities.
4. **Professional Registration**: Specialized portal for Architects/Engineers with real-time validation from ARB/ERB/SRB.
5. **Main Dashboard**: Role-based landing page showing quick actions and recent activity.

Applicant/Citizen & Professional Portal (8 Screens)

6. **Application Center**: Overview of all active and past permit applications.
7. **Submission Wizard**: Multi-step form for document uploads using the device camera or file system.
8. **GPS Site Mapper**: Dedicated screen to capture and verify GPS coordinates of the site.
9. **Status Tracker**: Real-time progress bar for current applications.
10. **Payment Portal**: Integration for fee calculation and PRN (Payment Reference Number) management.
11. **Review Hub**: For Professionals to review, link, and acknowledge client applications.
12. **Permit Validator**: QR code scanner for verifying building permits in the field.
13. **Anonymous Whistle Blowing**: Form for reporting illegal constructions with photo/GPS evidence.

Inspector/BCO & Committee Portal (7 Screens)

14. **Inspection Planner**: Calendar view of assigned field inspections.
15. **Digital Checklist**: Interactive forms for BCOs to complete technical reviews and supervision tasks on-site.
16. **Evidence Capture**: Full-screen camera interface for capturing GPS-tagged photos and notes.
17. **Stop Order Center**: Interface for issuing and managing Stop Order Notices.
18. **Committee Meeting Hub**: View scheduled meetings, agendas, and minutes.
19. **Decision Portal**: Interface for BCOs and Committee members to log approvals or denials.
20. **Reporting Dashboard**: High-level summaries for NBRB management.

3.6 possible AI enhancements (dependant on approval and availability of LLM models) – may be Phase 2

3.6.1. Computer Vision for Automated Compliance (BCO Feature)

Instead of a Building Control Officer (BCO) manually checking off every item, the app can use a **TensorFlow Lite** model (integrated into Flutter) to analyze photos in real-time.

- **Accessibility Detection**: When a BCO takes a photo of the building entrance, the AI identifies the presence of a **disability ramp** or **tactile paving**. If missing, the app flags a "Potential Non-Compliance" alert.
- **Infrastructure Recognition**: The AI can automatically detect and tag features like **lifts, fire extinguishers, or emergency exits** in uploaded photos.
- **Illegal Construction Detection**: By comparing the current photo with the **original approved drawings** , the AI can highlight structural deviations (e.g., an extra floor or unauthorized balcony).

3.6.2. Smart Form Autocomplete & OCR (Citizen/Professional Feature)

This feature targets the requirement for **"minimal data entry"** by learning from previous entries and external documents.

- **Intelligent Form Prefilling**: The app uses machine learning to predict and autocomplete fields (like "Building Type" or "Plot Location") based on the user's past 10 submissions or common patterns in that specific district.
- **OCR for Identity Documents**: Instead of typing in registration numbers, users can scan their **NIN, BRN, or TIN cards**. The app uses **Optical Character Recognition (OCR)** to extract the text and verify it against the backend in one tap.
- **Automated Drawing Extraction**: When a Professional uploads a drawing, the AI can scan the **title block** to automatically extract the "Project Title," "Architect Name," and "Plot Number," saving minutes of manual typing.

4. Project Duration Gantt Chart

Task / Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
PHASE 1: Inception & UI/UX Design												
Stakeholder Workshops												
UI/UX Wireframing (20+ Screens)												
Environment Setup (JIRA/Asana, PMT)												
Deliverable: Inception Report & Gantt Chart												
PHASE 2: Core Development – Professionals & Citizens												
Authentication Engine												
Submission Module (RQ-5)												
Geo-Tagging Engine												
Status Tracking (RQ-6)												
PHASE 3: Field Operations & BCO Tools												
Field Inspection Module (RQ-7)												
Enforcement Tools (RQ-9)												
Offline Synchronization (Hive/Sqflite)												
Notification System (FCM)												
PHASE 4: QA & Technical Review												
Internal Testing (Unit, Integration, SIT)												
Technical Reviews												
Security Audit (HTTPS/TLS, Auto-Logout)												
PHASE 5: UAT, Training & Deployment												
User Acceptance Testing (UAT)												
Training (NBRB Staff)												
Store Deployment (Play Store & Apple Store)												
Deliverable: UAT Certificate & Go-Live												

Phase 1: Inception & UI/UX Design (Weeks 1–2)

- **Stakeholder Workshops**: Conduct meetings to finalize the Statement of Understanding of the SRS.
- **UI/UX Wireframing**: Design 20+ screens in Figma for both Android and iOS, focusing on the "mobile-first" approach.
- **Environment Setup**: Configure the **JIRA/Asana** monitoring tool and co-opt the NBRB Project Management Team (PMT).
- **Deliverable**: **Inception Report & Gantt Chart** .

Phase 2: Core Development - Professionals & Citizens (Weeks 3–6)

- **Authentication Engine**: Implement NIN-based registration, Biometrics, and OTP-based 2FA.
- **Submission Module (RQ-5)**: Build the document vault for drawing uploads and link them to client applications.
- **Geo-Tagging Engine**: Integrate high-accuracy GPS capturing for site coordinates and anonymous whistle-blowing.
- **Status Tracking (RQ-6)**: Implement real-time status updates and URA PRN regeneration logic.

Phase 3: Field Operations & BCO Tools (Weeks 7–9)

- **Field Inspection Module (RQ-7)**: Develop digital checklists and GPS-tagged photo evidence capture for BCOs.
- **Enforcement Tools (RQ-9)**: Build the interface for issuing Stop Order Notices directly from the field.
- **Offline Synchronization**: Implement local caching (Hive/Sqflite) and background sync for intermittent connectivity.
- **Notification System**: Integrate Firebase Cloud Messaging (FCM) for push notifications and in-app messaging.

Phase 4: Quality Assurance & Technical Review (Weeks 10–11)

- **Internal Testing**: Conduct Unit, Integration, and System Integration Testing (SIT).
- **Technical Reviews**: Perform bi-weekly technical reviews and provide codebase access to the NBRB software team.
- **Security Audit**: Verify end-to-end encryption (HTTPS/TLS) and auto-logout functionality.

Phase 5: UAT, Training & Deployment (Week 12)

- **User Acceptance Testing (UAT)**: Execute UAT with stakeholders and obtain the signed UAT Certificate.
- **Training**: Conduct demonstration and training sessions for NBRB staff.
- **Store Deployment**: Deploy the finalized applications to the **Google Play Store** and **Apple Store**.
- **Deliverable**: **UAT Certificate & Go-Live** .

5. Financial Proposal

In accordance with the Terms of Reference (TOR) for the enhancement of the BIMS Mobile Application, we are pleased to submit our financial proposal. The total cost for the professional services described in the technical proposal is **UGX 15,000,000** (Fifteen Million Uganda Shillings Only).

Phase	**Activity**	**Amount (UGX**
Inception & UI/UX Design	Stakeholder Workshops (SRS Finalization)	
	UI/UX Wireframing :Design 20+ screens	
	Environment Setup (JIRA/Asana, PMT Onboarding)	
	Inception Report & Gantt Chart	2,000,000
Core Development – Professionals & Citizens	Authentication Engine (NIN, Biometrics, OTP 2FA)	
	Submission Module (RQ-5) Build the document vault for drawing uploads and link them to client applications.	
	Geo-Tagging Engine Integrate high-accuracy GPS capturing for site coordinates, Programmatically writing GPS coordinates (Latitude/Longitude) into the EXIF data of inspection photos. and anonymous whistle-blowing	6,500,000
	Status Tracking (RQ-6) Implement real-time status updates and URA PRN regeneration logic.	
	Permit Validation: A built-in QR code and Serial Number scanner for real-time validation of building permits	
Field Operations & BCO Tools	Field Inspection Module (RQ-7): Develop digital checklists and GPS-tagged photo evidence capture for BCOs. QR	
	Enforcement Tools (RQ-9): Build the interface for issuing Stop Order Notices directly from the field.	
	Offline Synchronization (Hive/Sqflite) Implement local caching (Hive/Sqflite) and background sync for intermittent connectivity.	
	Notification System (FCM Integration): Integrate Firebase Cloud Messaging (FCM) for push notifications and in-app messaging.	5,000,000
QA & Technical Review	Design and setup Unit tets, Integration & SIT Testing	
	Bi-Weekly Technical Reviews	
	Security Audit (HTTPS/TLS, Auto Logout): Verify end-to-end encryption (HTTPS/TLS) and auto-logout functionality.	1,000,000
UAT, Training & Deployment	User Acceptance Testing (UAT)	
	Training Sessions	
	Store Deployment (Google Play & Apple Store)	500,000
Total		**15,000,000**

6. Support and Maintenance

We shall offer support for a period of upto **6 months** after handover. The support will ber restricted to bug fixes, defects, and not feature requests or enhancements of the system. Any new feature requests or enhancements shall be billed separately.

The support will be offered as remote first approach. Any field support will incur a daily or hourly rate to be negotiated.

7. Declaration

We confirm that we will provide high-performance, scalable mobile apps and handle the entire lifecycle—from design to support —at no additional cost for any defects identified during the DLP.

Name: Byaruhanga Raymond

Signature: _____ _____ **Date**: _____19th February 2026_____